A Semi-Supervised Domain Alignment Transformer for Hyperspectral Images Change Detection

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***Abstract*— Supervised deep learning (DL)-based hyperspectral images change detection (HSIs-CD) has demonstrated excellent performance; however, current methods require many labeled training samples, and labeling the dataset is labor-intensive, limiting the application of high-precision supervised learning. Besides, there has been a lack of breakthroughs in unsupervised HSIs change detection (CD) methods due to the different feature distributions of bitemporal HSIs. Here, we propose a semi-supervised domain alignment transformer (DA-Former) for HSIs-CD to address the issues with limited samples. Specifically, a dual-branch transformer autoencoder (TAE) is designed, where the middle layer weights of the dual-branch transformer are shared, pulling features from different data into the same space. Moreover, the TAE is also trained cyclically to align the domains. Although the bitemporal HSIs features are cross domain, there is still confusion between the features of different objects. Thus, two fully connected (FC) layers are employed to classify the HSIs middle features extracted by the TAE into changed class or unchanged class with limited labeled data. Three HSIs-CD datasets are used to test this method, showing that TAE can align bitemporal HSIs domains and achieve the highest accuracy compared with benchmark approaches. The code of the proposed method will be published at https://github.com/yanhengwang-heu/IEEE_TGRS_DA-Former.**

***Index Terms*— Autoencoder, change detection (CD), domain alignment, hyperspectral images (HSIs), transformer.**

I. Introduction

SATELLITE sensors monitor the Earth's surface periodically, making it possible to detect its changes. Change detection (CD) focuses on extracting the different areas of the same region over different times in remotely sensed imagery. The CD process has underpinned many practical applications in land regulation, urban planning, and natural hazard assessment. Hyperspectral images (HSIs) can be imaged continuously across the ultraviolet to mid-infrared electron spectrum, enabling CD information detection at fine spectral scales. The hyperspectral images CD (HSIs-CD) has become a hot research topic in the remote sensing community.

HSIs have more bands than multispectral images (MSIs), which provide richer spectral information but also information redundancy and subtle changes. Several HSIs-CD methods have been developed and widely used. For example, based on algorithmic evolution, HSIs-CD methods can be divided into algebra-based, traditional machine learning (TML)-based, and deep learning (DL)-based. Algebra-based methods obtain the direct distance between two temporal HSIs and then use thresholds to classify the distances into changed and unchanged classes. For example, change vector analysis (CVA) and compressed CVA (C^2VA) [1] are classic algebra-based CD algorithms for both MSIs and HSIs. Liu et al. [2] designed a sequential spectral CVA (S^2CVA) for HSIs-CD only to analyze subtle changes. The thresholds were often set via trial-and-error, while TML was used later on for automatic thresholding, such as expectation maximization (EM) [3]. TML-based methods transform the HSIs into features and classify them as changed or unchanged. DL-based methods have more parameters, all updated by the task automatically without human intervention. Moreover, the DL activation function provides nonlinear feature extraction, further increasing the representation capability [4]. DL-based methods quickly become popular for HSIs classification, as well as HSIs unmixing [5], [6], [7].

HSIs-CD methods can also be divided into unsupervised, semi-supervised, and supervised learning methods in relation to the amount of training samples. Unsupervised methods involve tasks performed directly without labeled data. Semi-supervised methods use unsupervised algorithms to learn features from unlabeled data and then fine-tune the algorithm using a small amount of labeled data, whereas supervised methods only use the labeled data to train a model. For unsupervised HSIs-CD methods, typical methods are algebra-based, TML-based, and DL-based. The algebra-based methods include Euclidean distance (ED), the absolute average difference (AAD), vector angle (VA), and CVA [8], among others. TML-based HSIs-CD methods typically include principal

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component analysis (PCA) [9], multivariate alteration detection (MAD) [10], iteratively reweighted MAD (IR-MAD) [11], and K-means [12]. K-means also can be used for automatic thresholding. DL-based unsupervised methods include the autoencoder, generative adversarial network (GAN), and pseudo-label approaches. The autoencoder is a neural network model that maps the input data to itself, within which the encoder maps the input to low-dimensional features as latent variables, and the decoder recovers the low-dimensional features as input. The low-dimensional features are used for classification or CD. For example, Liu et al. [13] used a transformer as the backbone of the autoencoder to extract the low-dimensional features that can be used for CD and edge detection. Hu et al. [14] deployed a Siamese autoencoder to detect anomaly change. Given that GANs comprise a generator and a discriminator, the generator transforms one distribution into another, and the discriminator determines how well the generated distribution. Lei et al. [15] treated the encoder as a generator and pulled low-dimensional features into a particular distribution via a discriminator. The pseudo-label method is an algorithm that uses algebra-based and TML-based unsupervised learning methods to obtain CD results, from which high-quality results are found as labeled datasets to train supervised DL methods, such as high confidence results of CVA were used as the labeled samples [16]. Li et al. [17] integrated the CVA, PCA-CVA, and IR-MAD results to obtain high-confidence labeled samples. Liu et al. [18] obtained number of changes and multiclass change separation through unsupervised adaptive spectral change vector representation (ASCVR). Pseudo-labels were selected and fed into a supervised algorithm to increase the accuracy of multiclass CD. Du et al. [19] obtained the high-confidence labeled samples by CVA and used SVM and posterior probability comparison to achieve the tri-temporal images CD results. Recently, unsupervised learning has evolved significantly with increasing accuracy, but still, there has not been a major breakthrough yet.

For semi-supervised HSIs-CD methods, little research has been done using algebra and TML [20], [21]. S^2CVA is a semi-supervised method for multiclass HSIs-CD [2]. S^2CVA obtains binary CD results by C^2VA. It then uses ASCVR to obtain multiclass CD results and iteratively achieves highly accurate results through manual interaction. Yuan et al. [22] exploited Laplacian Regularized Metric Learning with many unlabeled and limited labeled data. Additionally, Liu et al. [23] used a large amount of unlabeled data to obtain high-quality pseudo-labels on the existing CD algorithm and then used the pseudo-labels and the small amount of labeled data to train the CD algorithm together. Both studies are based on TML algorithms with limited feature extraction capabilities.

For supervised HSI-CD methods, the mainstream approaches in the literature are TML and DL. Common TML supervised methods include support vector machine (SVM) [24], [25], random forest (RF) [26], spectral angle mapper (SAM), and maximum likelihood classifier (MLC) [8]. Over the past decade, DL has transformed this field with strong nonlinear feature representations, including convolutional neural networks (CNNs), attention mechanisms, recurrent neural networks (RNNs), graph neural networks (GCNs), and transformers. CNNs demonstrated strong capabilities to extract spatial features with convolutional kernels sliding over the image. For instance, Ou et al. [27] proposed a CNN framework to select bands and fuse features for HSIs-CD. Wang et al. [28] used a CNN to detect HSIs change with different spatial resolutions. Yang et al. [29] used a depth-wise separable convolution to extract the multiscale features. An unsupervised autoencoder was developed to extract spectral-spatial features from HSIs using 3D-CNN, followed by a supervised 2D-CNN to classify the features for CD [30]. Attention mechanisms are designed to identify the features of the input data that are important to the task by designing the key, query, and value structure. For HSIs-CD, various attention mechanisms have been designed to extract HSI features. For example, Wang et al. [31], [32] proposed spectral-spatial attention to find the spectral bands and spatial locations of HSIs with high contribution to CD. Chen and Shi [33] offered spatial-temporal attention to determine the weight of any pixel of bitemporal HSIs, which resulted in more discriminative features. RNNs are the structure that learns the sequence information of data. They are used to receive the sequence information of the multitemporal HSIs for evaluating the changes. Long short-term memory (LSTM), a type of RNN, was designed to analyze the temporal sequence information between multitemporal HSIs [34]. GCNs extend the convolutional operations from gridded data to a graph representation, which can learn unstructured image information for HSIs-CD applications. Wang et al. [35] proposed a spatial-temporal graph attention to learn the non-Euclidean space information with more important bands and spatial locations. Qu et al. [36] made a dual-branch GCN to extract the non-Euclidean spatial features and a GCN layer to fuse the features of bitemporal HSIs. A Transformer is capable of learning sequences (spatial, spectral and temporal) and non-Euclidean spatial information on these sequences. Wang et al. [37] proposed a spectral-spatial-temporal transformer (SST-Former), where a dual-branch spectral and spatial transformer was used to extract the bitemporal HSI features, and a temporal transformer was used to compare the features. Hong et al. [38] rethought the relationship between bands and exploited a transformer to extract detailed features in HSIs. Despite the high accuracy achieved in current DL-based supervised CD methods, a significant amount of training data is required to learn deep networks sufficiently well.

MSIs focus on the extraction of color, texture, shape, and spectral features, whereas HSIs CD methods focus more on the extraction of spectral features of patches. Because of different imaging conditions, the spectral values and curves of the same objects or different objects are different on bitemporal HSIs, and the spectral features also have some differences under the same feature extractor; therefore, we can treat bitemporal HSIs as different image domains. The spectral values of different objects may have a small difference, and even the different object has the same spectra. So, these domains between different features could overlap, similar to the source domain of the bitemporal images shown in Fig. 1. Triangles and

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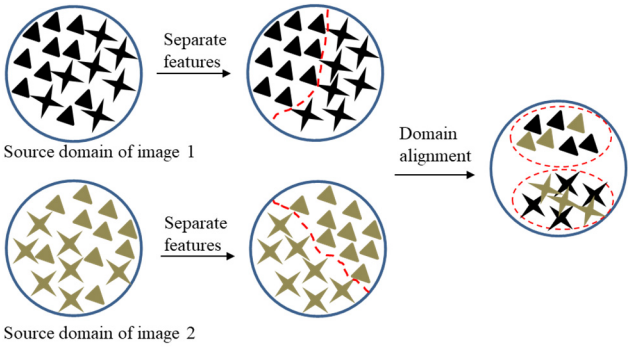


Fig. 1. Illustration of the domain alignment process.

stars represent the features of the two objects, respectively, and different colors represent different values. Supervised methods can separate the overlapping features easily and align the domain between the temporal images using massive training data during learning, which often demonstrates excellent performance in CD. The automatic domain alignment process in supervised learning is illustrated in Fig. 1. As for unsupervised approaches, they do not separate the overlapping features well and align the feature domains of bitemporal HSIs. They typically compare the features directly in different domains. Hence, the areas of overlapping features and different domains are still hard-to-detect. In semi-supervised methods, the features are extracted by unsupervised approaches, and a few labeled data are used to detect changes in features; however, the features extracted by unsupervised approaches are on different domains, and few labeled samples have limited automatic domain alignment and feature separation capabilities, and are also prone to overfitting. Besides, current studies focus on TML only, which has limited feature extraction capability for complex HSIs.

In this article, we propose a semi-supervised domain alignment transformer (DA-Former) for HSIs-CD, to reduce the labeling processes and improve feature extraction capabilities for HSIs. A large amount of unlabeled data is used to achieve domain alignment through a dual-branch autoencoder, and a small amount of labeled data is used to separate different features and train the classifier. The main contribution of the proposed DA-Former can be summarized as follows.

1) A transformer autoencoder (TAE) is proposed for HSIs, which unsupervised extracts the spatial and spectral sequence information to fully extract features to solve the problem of information redundancy and rich subtle changes in HSIs.
2) A dual-branch TAE, comprising two TAE, is for unsupervised feature extraction of bitemporal HSIs. The middle layers of the dual branch have shared weights to align the feature domain. Additionally, a cycle TAE is designed for additional domain alignment.
3) A small amount of labeled data and a weighted loss function are used to separate the features of different objects and train the fully connected (FC) layers to classify the middle features of the bitemporal HSIs into a changed or unchanged class. Such a strategy can address the issue of overfitting.

II. Proposed DA-Former Method

In this section, Part A introduces TAE's network structure, followed by the domain alignment in Part B. The classifier is presented in Part C, and Part D describes the loss function. Fig. 2 shows the flowchart of the proposed DA-Former architecture in detail.

A. Transformer Autoencoder

The autoencoder obtains the latent low-dimensional features of the input by reconstructing the input. The autoencoder can be divided into the encoder $e(x)$ and the decoder $d(x)$. The reconstruction process is presented as follows:

$$\text{out} = d(e(x)). \tag{1}$$

The reconstruction loss can be derived by

$$L_r = \frac{1}{whc}\sum_{i=1}^{whc}(x - \text{out}) \tag{2}$$

where w, h, and c are the width, height, and channel of input x, respectively.

Autoencoders are typically built using multilayer perceptron (MLP) and CNN. In this article, we build an autoencoder using a transformer (TAE) to extract the spatial and spectral sequence information, which can fully extract the features to solve the problem of information redundancy and subtle change in HSIs. Specifically, a spectral-spatial transformer (SS-Former) [37] is used as TAE's encoder $e(x)$. SS-Former consists of a linear projection (LP) layer of dimension m, an LP layer of dimension n, a position embedding, two spectral transformer encoders, three spatial transformer encoders and a class token. Two LPs are used to reduce the spectral-spatial dimension and information redundancy of HSIs. Spectral transformer encoders are used to extract spectral sequence information of HSIs. Spatial transformer encoders are used to extract spatial sequence information of HSIs. Wang et al. [37] described further details on the SS-Former. We remove the class token employed in [37] mathematically considering $x \in \mathbb{R}^{wh\times c}$ as an HSI patch. After the SS-Former, a middle feature f, $f \in \mathbb{R}^{m\times n}$ is acquired, where m and n are the spectral and spatial LP dimensions, respectively.

For the decoder, we first embed f into a layer normalized (LN) and a multihead self-attention (MSA) to model the sequential relationship between the elements within f. An LP is used to recover the spatial dimension, which is calculated as follows:

$$z_1 = f + \text{MSA}(\text{LN}(f)), \quad z_1 \in \mathbb{R}^{m\times n} \tag{3}$$

$$z_2 = \omega_1 z_1, \quad z_2 \in \mathbb{R}^{m\times wh} \tag{4}$$

where z_i is the output of decoder layer i, and ω_1 is the weight of LP. We then reshape $z_2 \in \mathbb{R}^{m\times wh}$ into $z_2 \in \mathbb{R}^{wh\times m}$ and use an LN and an MLP for non-LP as follows:

$$z_3 = z_2 + \text{MLP}(\text{LN}(z_2)), \quad z_3 \in \mathbb{R}^{wh\times m}. \tag{5}$$

Finally, an LN and an LP are used to transform the channel of z_3 from m to c

$$out = \omega_2 LN(z_3), \quad \text{out} \in \mathbb{R}^{wh\times c} \tag{6}$$

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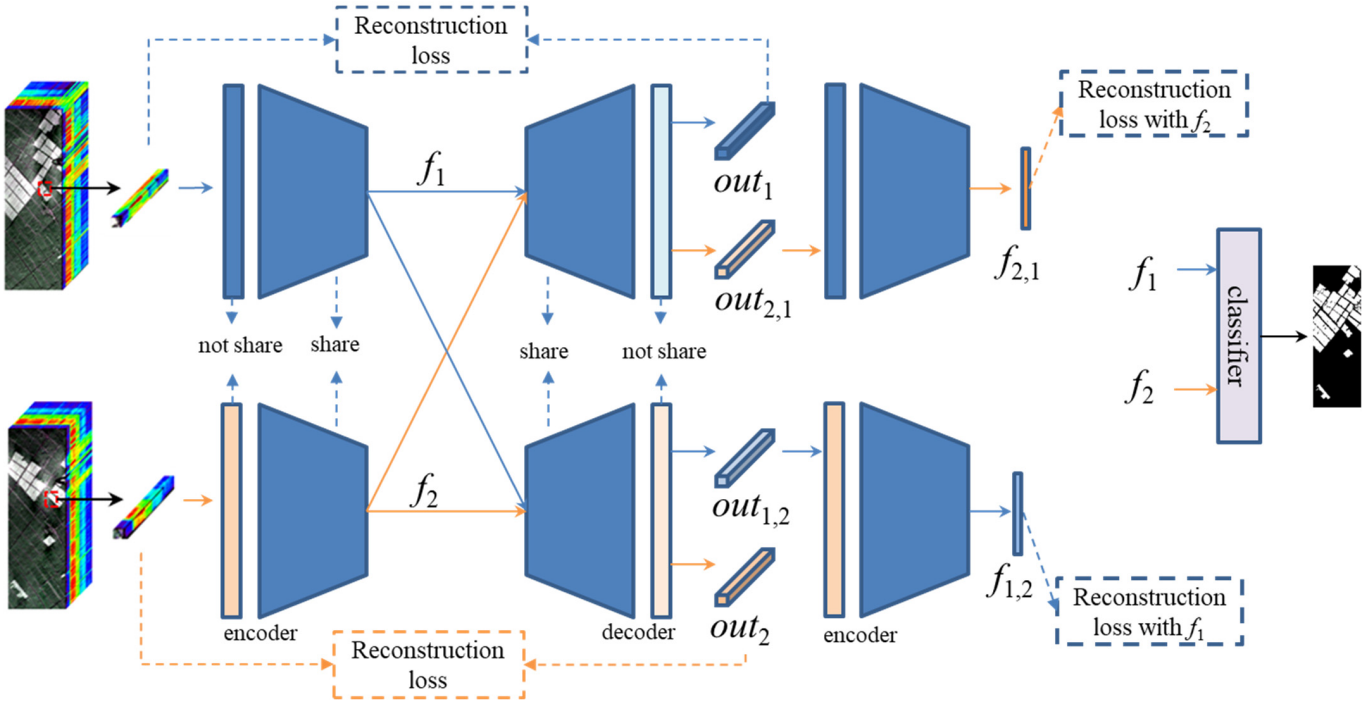


Fig. 2. Framework of the proposed DA-Former for HSIs-CD, comprising a dual-TAE with a cycle structure and a classifier. The blue trapezoids indicate the encoder and decoder parts with shared parameters, and the rectangles with different colors indicate the encoder and decoder parts with nonshared parameters (same colors and shapes indicate the same parameters and structures).

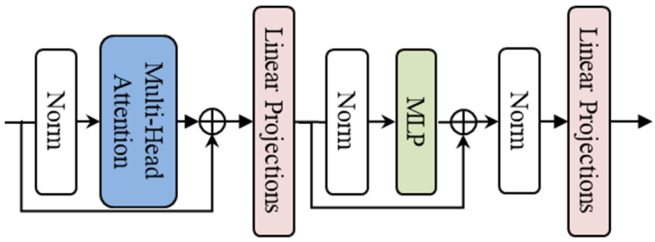


Fig. 3. Structure of the decoder.

where out is the reconstruction of x in the TAE. Fig. 3 demonstrates the decoder's structure.

For bitemporal HSIs, two inputs x_1 and x_2, are fed into two autoencoders, and two outputs, out_1 and out_2, are provided. The two autoencoders have a dual-branch TAE for HSIs-CD, as illustrated in Fig. 2.

B. Domain Alignment

The same objects on different temporal HSIs will have different spectral features obtained under the same TAE, and different objects may have the same features obtained under the same TAE. It is difficult for unsupervised feature comparison. So, the dual-branch TAE is designed to achieve unsupervised domain alignment using the prior knowledge that they belong to different domains. This domain alignment TAE is named as DA-Former.

Within the DA-Former, the weight of the first LP layers and the last LP layers are unshared, and the other layers are shared. The first LP layers can map the features of the bitemporal HSIs onto the same place, the features of the middle layers are in the same domain, and the last LP layers allow the features in the target domain to be recovered into source domains.

Moreover, a cycle TAE is an important part of the DA-Former for further domain alignment. Given the two encoders and two decoders of dual-branch TAE as $e_1(x)$, $e_2(x)$, and $d_1(x)$, $d_2(x)$. We obtain two middle features f_1, f_2 and the reconstruction output out_1, out_2 by

$$f_1 = e_1(x_1) \tag{7}$$

$$f_2 = e_2(x_2) \tag{8}$$

$$\text{out}_1 = d_1(f_1) \tag{9}$$

$$\text{out}_2 = d_2(f_2). \tag{10}$$

Otherwise, we use $d_2(x)$ to decode the f_1 and obtain the reconstruction x_1 with style x_2, as follows:

$$\text{out}_{1,2} = d_2(f_1) \tag{11}$$

where $\text{out}_{1,2}$ is the reconstruction x_1 with style x_2. The $\text{out}_{1,2}$ is thereafter put into the $e_2(x)$ and obtain a feature $f_{1,2}$ by

$$f_{1,2} = e_2(\text{out}_{1,2}). \tag{12}$$

The smaller the distance loss between f_1 and $f_{1,2}$, the better the domain alignment.

At the same time, we obtain $\text{out}_{2,1}$, which is a reconstruction x_2 with style x_1 and a feature $f_{2,1}$ that computes the loss based on f_2. The two losses are the cycle TAE loss, which is used to align the bitemporal HSI features domain.

C. Classifier

The feature domains of the bitemporal HSIs are aligned, but the hard-to-detect area features also overlap, such as the area near the red dotted line in the aligned domain in Fig. 4. We use a few labeled data to train the dual-branch TAE to separate the overlapping features, thus achieving the target

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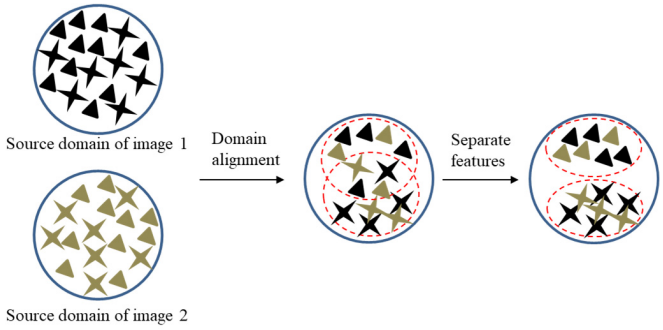


Fig. 4. DA-Former domain alignment and feature separation process.

domain. The domain alignment and feature separation process of DA-Former can be seen in Fig. 4.

At the same time, we subtract their middle features, f_1 and f_2, and then use two FC layers to classify them into changed or unchanged classes. The CD result is noted as y'. Since f_1 and f_2 are in the same domain, a small number of labeled samples can train a robust classifier without overfitting.

D. Loss Function

Five losses are used to update the weight of DA-Former. L_{r1} and L_{r2} are reconstruction losses of bitemporal HSIs, and the reconstruction loss is presented in (2). Besides, the mean square error (MSE) of f_1 and $f_{1,2}$ and the MSE of f_2 and $f_{2,1}$, as the domain alignment losses, are calculated by

$$L_{da1} = \frac{1}{mn}\sum_{i=1}^{mn}(f_1 - f_{1,2}) \tag{13}$$

$$L_{da2} = \frac{1}{mn}\sum_{i=1}^{mn}(f_2 - f_{2,1}). \tag{14}$$

The final loss used is the CD loss, which can be formulated as follows:

$$L_{cd} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log\left(y_i'\right) + (1 - y_i)\log\left(1 - y_i'\right)\right]. \tag{15}$$

The loss of DA-Former is derived from

$$L = L_{r1} + L_{r2} + L_{da1} + L_{da2} + \lambda L_{cd} \tag{16}$$

where $\lambda \in [0, 1]$ is the weight of L_{cd}. Since L_{cd} is acquired using few samples, a small λ can reduce the overfitting of DA-Former.

III. Experiments

A. Datasets

The proposed DA-Former was tested on three HSI-CD datasets: the Farmland,[1] the Santa Barbara, and the Bay Area datasets.[2]

[1]https://rslab.ut.ac.ir/data

[2]https://citius.usc.es/investigacion/datasets/hyperspectral-change-detection-dataset

1) Farmland Dataset: This dataset was provided by the Remote Sensing Laboratory, School of Surveying and Geospatial Engineering, College of Engineering, University of Tehran. The bitemporal HSIs were acquired using the Hyperion hyperspectral sensor on May 3, 2006 and April 23, 2007, respectively. The location is farmland near Yuncheng, Jiangsu province, China. There are 242 bands covering the spectral range 0.4–2.5 μm in this dataset. A total of 154 spectral bands are used for CD after removing noise and water absorption bands. These include bands 8–57 (426.82–925.41 nm), bands 82–119 (962.91–1336.15 nm), bands 134–164 (1487.53–1790.19 nm), bands 183–184 (1981.86–1991.96 nm), bands 188–220 (2032.35–2355.21 nm). The size of the two HSIs is 420 × 140 pixels with a spatial resolution of 30 m. There are 40 417 unchanged pixels and 18 383 changed pixels. This article used only 100 unchanged and 100 changed pixels as the training data. The pseudo-color images (R: 752.43, G: 640.5, B: 528.57 nm) are depicted in Fig. 5(a) and (b).

2) Santa Barbara Dataset: This dataset was provided by the CiTIUS, Universidad de Santiago de Compostela, Spain. The bitemporal HSIs were acquired using the airborne visible infrared imaging spectrometer (AVIRIS) sensor in 2013 and 2014, covering an area of Santa Barbara in California, USA. Each image has a size of 984 × 740 × 224 pixels ($w \times h \times c$) over the 0.4–2.5 um spectral range with a 20 m/pixel spatial resolution. There are 80 418 unchanged pixels, 52 134 changed pixels, and 595 608 unknown pixels. Only 100 unchanged and 100 changed pixels were used to train the DA-Former. The pseudo-color images (R: 685.19, G: 546.96, B: 438.76 nm) are illustrated in Fig. 5(c) and (d).

3) Bay Area Dataset: This dataset was also provided by the CiTIUS, Universidad de Santiago de Compostela, Spain. The bitemporal HSIs were acquired using the AVIRIS sensor in 2013 and 2014, respectively. The covered land is in Patterson, California, USA, having a size of 600 × 500 × 224 pixels ($w \times h \times c$) over the 0.4–2.5 um spectral range. The spatial resolution is 20 m/pixel. There are 34 211 unchanged pixels, 38 425 changed pixels, and 227 364 unknown pixels. Only 100 unchanged and 100 changed pixels were used as the training data. The pseudo-color images (R: 625.95, G: 556.83, B: 468.18 nm) are presented in Fig. 5(e) and (f).

B. Experimental Setup

The overall accuracy (OA) and kappa coefficient (Kappa) were used to assess the CD performance. In addition, a confusion matrix was established to acquire the true positive (TP), true negative (TN), false positive (FP), and false negative (FN), where TP indicates the number of changed pixels in predicted results and ground truth, TN is the number of unchanged pixels in the predicted result and ground truth, FP is the number of unchanged pixels in the ground truth while changed in the predicted result, and FN is the number of unchanged pixels in the predicted result while changed in the ground truth. FN is the opposite of FP. The OA is calculated by

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{FP} + \mathrm{TN} + \mathrm{FN}}. \tag{17}$$

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(a) (b) (c) (d) (e) (f)

Fig. 5. HSIs-CD datasets. (a) Farmland dataset on 3 May 2006. (b) Farmland dataset on 23 April 2007. (c) Santa Barbara dataset in 2013. (d) Santa Barbara dataset in 2014. (e) Bay Area dataset in 2013. (f) Bay Area dataset in 2014.

TABLE I

ACCURACY COMPARISON OF EXCELLENT HSI-CD METHODS ON THREE DATASETS

		CVA	SVM	SSDM	Re3FCN	CSANet	SST-Former	DA-Former
Farmland	Kappa	0.6998	0.7519	0.7846	0.7828	0.8176	0.8288	**0.8721**
	OA	0.8749	0.8891	0.913	0.9091	0.9190	0.9233	**0.9436**
Santa Barbara	Kappa	0.6519	0.9002	0.7158	0.8985	**0.9532**	0.9291	0.9495
	OA	0.8325	0.9517	0.8649	0.9516	**0.9778**	0.9665	0.9759
Bay Area	Kappa	0.7063	0.7188	0.7937	0.7870	0.9376	0.9172	**0.9421**
	OA	0.8517	0.8599	0.9015	0.8935	0.9690	0.9587	**0.9712**

Bold fonts are best and blue fonts are 2nd-best.

For images with a few changed pixels, TN and TP are large regardless if a change is detected. Since OA cannot evaluate the results effectively, Kappa is used to balance the evaluation of different classes of results. The Kappa metric is calculated by

$$\text{Kappa} = \frac{\text{OA} - p_e}{1 - p_e} \tag{18}$$

$$p_e = \frac{(\text{TP}+\text{FP})(\text{TP}+\text{FN})(\text{TN}+\text{FN})(\text{TN}+\text{FP})}{(\text{TP}+\text{FP}+\text{TN}+\text{FN})^2}. \tag{19}$$

The DA-Former was trained using PyTorch 1.8.1 + cu111 on an Intel Xeon CPU and an NVIDIA GeForce RTX 3060 12G GPU. The optimizer was Adam with default parameters. The learning rate was $5e$-4, and at every 30 epochs, the learning rate was multiplied by 0.95. The total epochs were 600. Gaussian error linear unit (GeLU) was used as the activation function in TAE, and rectified linear unit (ReLU) was used as the activation function in the classifier. For the hyperparameters of SS-Former, please refer to [37]. The m and n were set to 64. The patch and batch sizes were 9 and 64, respectively, and $\lambda = 0.4$. The number of unlabeled patch samples was 2000. The number of labeled samples for each class on three datasets was 100. As the detection range expanded, the training samples can be increased appropriately.

Six state-of-the-art CD methods were used as benchmark approaches to compare with DA-Former, including CVA, SVM, semi-supervised distance metric (SSDM) [22], recurrent 3-D fully convolutional network (Re3FCN) [39], cross-temporal interaction symmetric attention network (CSANet) [40], and SST-Former [37].

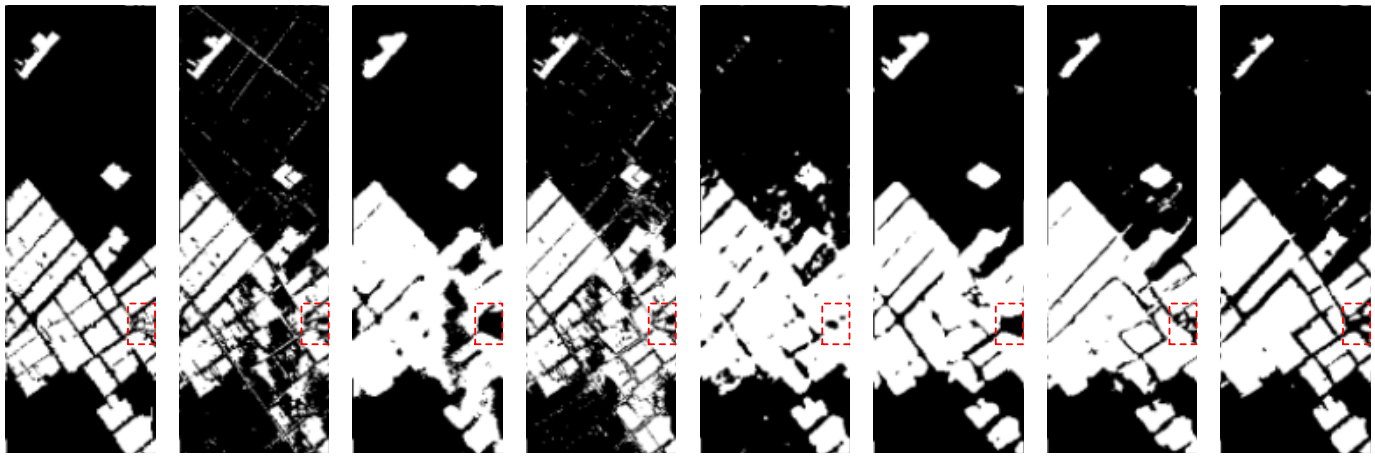
C. Experimental Results

We demonstrate benchmark comparison results both visually and quantitatively. All methods were trained thrice, and their average performance is reported in Table I. Furthermore, the visible results of Farmland, Santa Barbara, and Bay Area are illustrated in Figs. 6–8, respectively.

For the Farmland dataset, CVA has the lowest OA and Kappa, and the visual result is unsatisfactory. The SVM has the second lowest OA and Kappa, and many alarm areas are missing in the visual results. The accuracy of SSDM and Re3FCN is higher than SVM, but still, it is not an appealing result. Furthermore, for visual results, SVM also has missing alarm areas. The Re3FCN can easily be affected by the unchanged area around the changed area and create a false alarm. The CSANet and SST-Former further improve the HSIs-CD accuracy, which is partially acceptable. Compared to the second-best SST-Former, the OA and Kappa of the DA-Former increased by 2% and 5%, respectively. Fig. 6(f)–(g) infers that CSANet, SST-Former, and DA-Former are the best-performing algorithms that outperform all benchmarks. Regarding the hard-to-detect region within the red dashed box, DA-Former and SST-Former can detect it, but CSANet cannot. DA-Former has a better OA and Kappa than CSANet; however, their poor detection performance of the change region boundary is caused by a large patch.

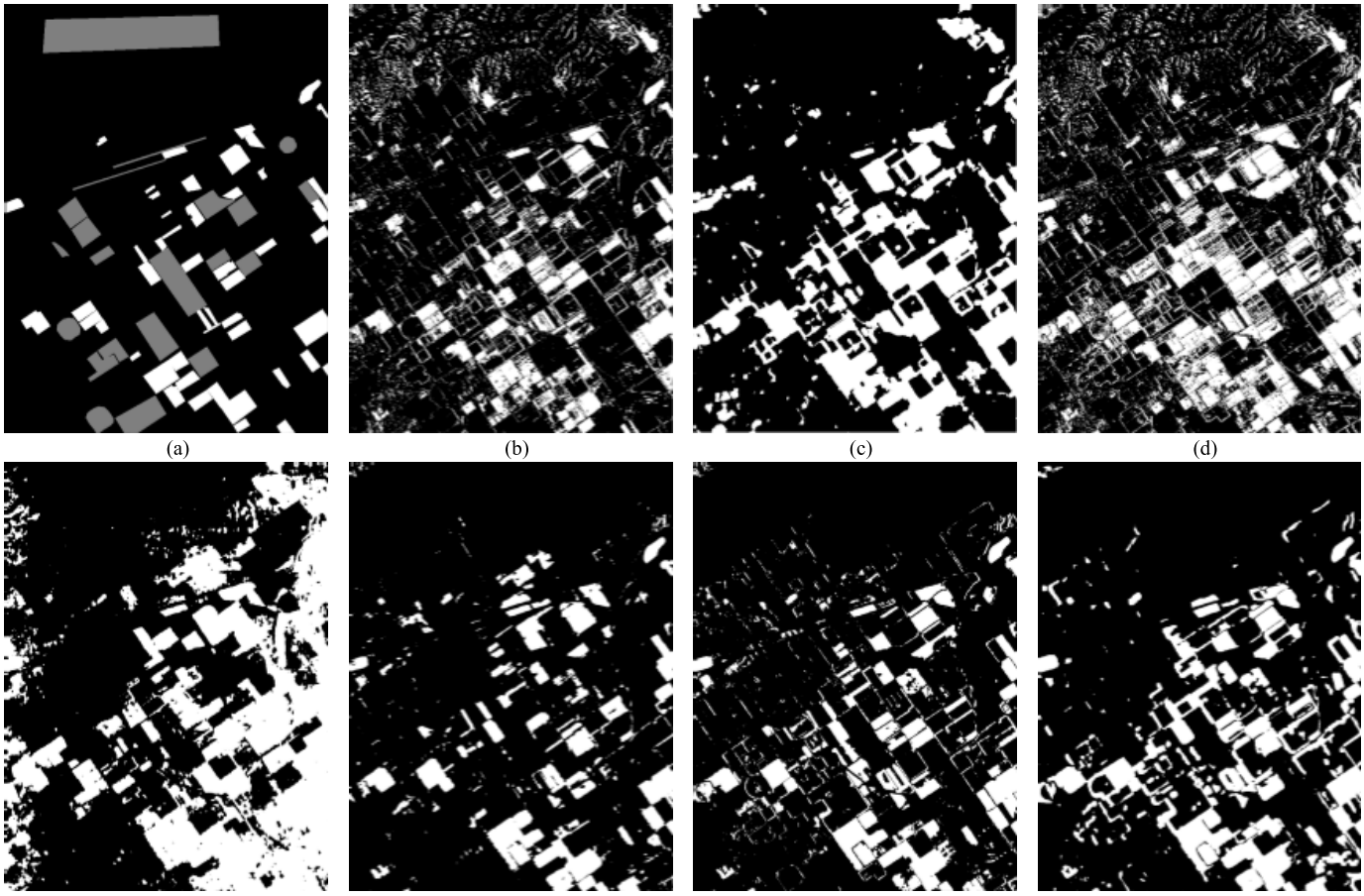
For the Santa Barbara dataset, CVA has the lowest accuracy. There are many false alarm areas and sporadic salt and pepper noise in Fig. 7(b). SSDM has a small improvement over CVA. In Fig. 7(d), no large false or missing alarm areas compared with ground truth are found. SSDM's low accuracy

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(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 6. CD results of different methods on the farmland dataset. (a) Ground truth. (b) CVA. (c) SVM. (d) SSDM. (e) Re3FCN. (f) CSANet. (g) SST-Former. (h) DA-Former. Changed pixels are white, and unchanged pixels are black. The red dotted box means that the area is hard-to-detect.



(a) (b) (c) (d)

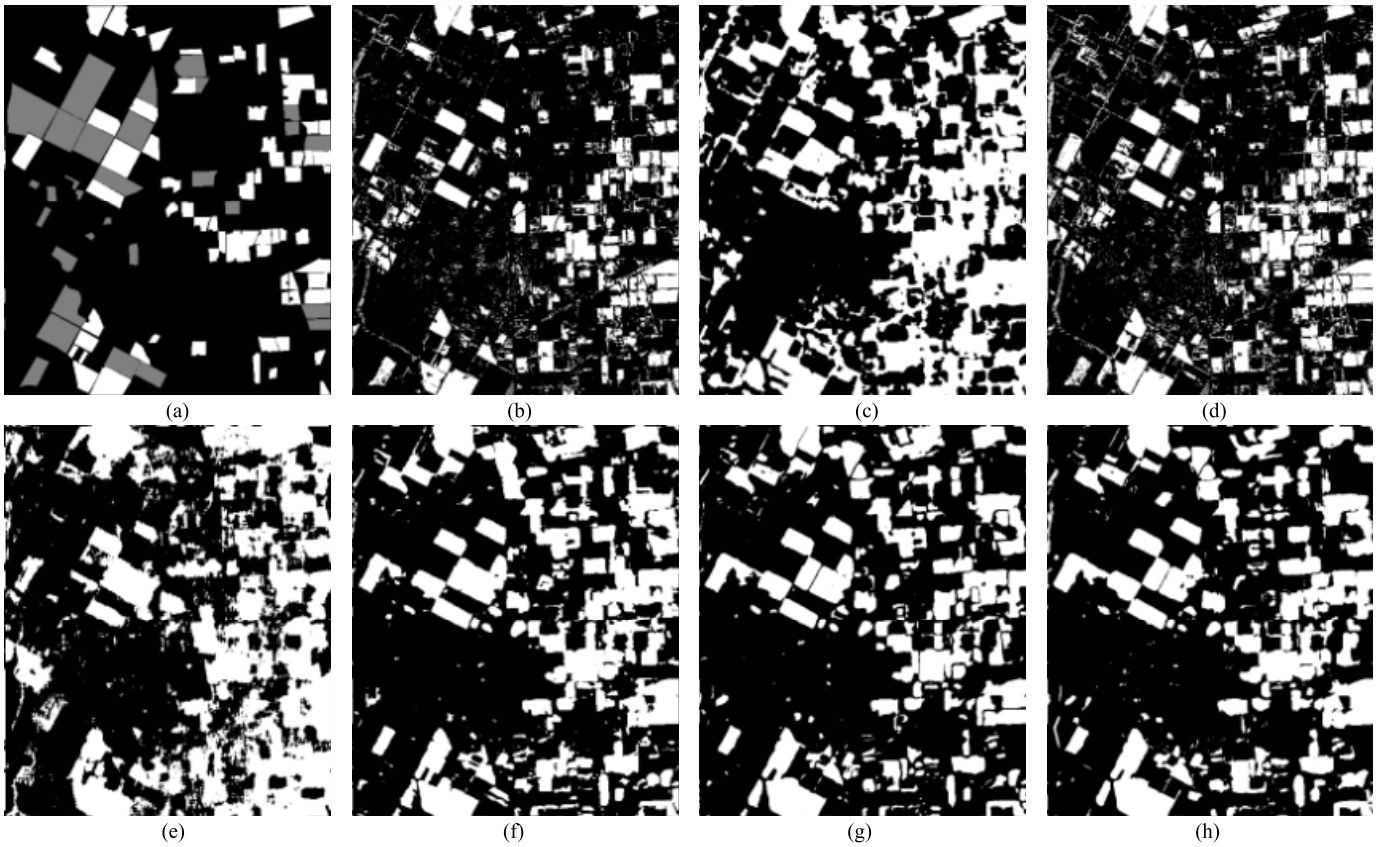
(e) (f) (g) (h)

Fig. 7. CD results of different methods on the Santa Barbara dataset. (a) Ground truth. (b) CVA. (c) SVM. (d) SSDM. (e) Re3FCN. (f) CSANet. (g) SST-Former. (h) DA-Former. Changed pixels are white, unchanged pixels are gray, and unknown pixels are black.

is the result of excessive sporadic salt and pepper noise. SVM and Re3FCN outperform SSDM by approximately 19% on the Kappa. SVM and Re3FCN also have false alarm areas shown in Fig. 7(c) and (e). Furthermore, SST-Former presents an increased OA and Kappa and obtains an appealing visual result. The highest accuracy is presented by CSANet, and the second highest accuracy is achieved by DA-Former, which presented OA and Kappa difference of just 0.19% and 0.37%, respectively; however, there are a large number of boundary detection errors in the CSANet visual result. The DA-Former visual result is more excellent than others, with only a few missing alarm and false alarm areas.

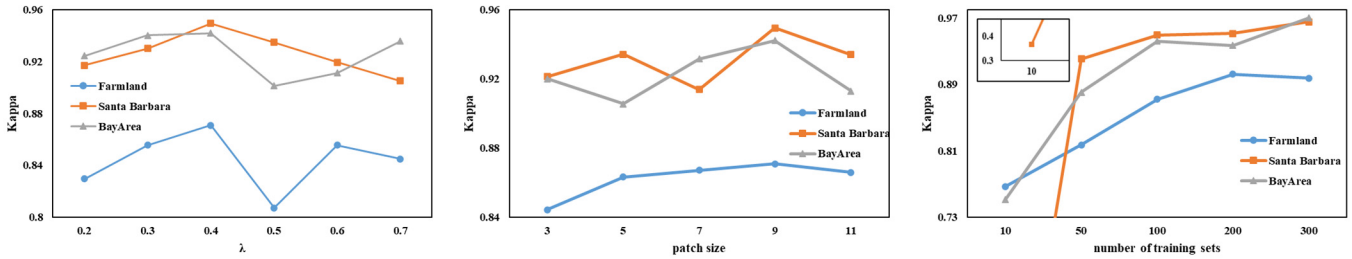
For the Bay Area dataset, regarding CVA and SVM, they both have about 70% kappa and 85% OA. The Kappa of SSDM and Re3FCN are 7.49% and 6.82% higher than SVM, which is not considered a high precision for supervised and semi-supervised algorithms. The visual results of SVM and SSDM have some wrong detection areas and sporadic salt

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(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 8. CD results of different methods on the Bay Area dataset. (a) Ground truth. (b) CVA. (c) SVM. (d) SSDM. (e) Re3FCN. (f) CSANet. (g) SST-Former. (h) DA-Former. Changed pixels are white, unchanged pixels are gray, and unknown pixels are black.



(a) (b) (c)

Fig. 9. Kappa of different hyperparameters on three datasets. (a) L_{cd} weight λ. (b) Patch size. (c) Number of training samples.

and pepper noise in Fig. 8(b) and (d). The boundaries of the objects in the visual results of SVM and Re3FCN are unclear [Fig. 8(c) and (d)]. Compared to Re3FCN, SST-Former and CSANet presented an increased Kappa that exceeds 10%. CSANet achieves the second-highest accuracy. Although SST-Former and CSANet perform well [Fig. 8(f) and (g)], some small false and missing areas are still present.

Overall, DA-Former has the highest OA and Kappa, and according to Fig. 8(h), DA-Former yields less mis-detected areas than other methods. Clearly, CVA is one of the best approaches for unsupervised tasks where accuracy is not a major requirement. SVM and Re3FCN further increase the accuracy of HSIs-CD but still fall short of the standard. SSDM does not perform well with a sample size of 100. CSANet and SST-Former achieve better results, but their performance is not stable across different datasets. DA-Former further increases the accuracy of HSIs-CD with limited samples and performs consistently on different datasets. The accuracy of DA-Former is not far from supervised learning on big data. The above experiments have demonstrated that the semi-supervised DA-Former algorithm is effective.

D. Ablation Analysis

With limited samples, the function of the semi-supervised DA-Former relies mainly on the unsupervised features of the TAE and domain alignment learning (dual-TAE middle layer parameter sharing and cycle TAE), all of which are analyzed below through ablation for their effectiveness. We consider five experiments on the Farmland dataset, including transformer encoder + classifier (Base), TAE + classifier (TAE-C), dual-TAE with shared middle layers parameters + classifier (Shared TAE-C), cycle TAE + classifier (Cycle TAE-C) and DA-Former, and the results are shown in Table II.

By comparing the accuracy of the base and TAE-C, we find that the unsupervised features provided by the TAE can be applied to CD and increase the accuracy of CD under limited

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TABLE II
ABLATION ANALYSIS OF DA-FORMER ON FARMLAND DATASETS

Methods	TAE	Shared TAE	Cycle TAE	OA	Kappa
Base	×	×	×	0.9118	0.8008
TAE-C	✓	×	×	0.9208	0.8215
Shared TAE-C	✓	✓	×	0.9267	0.8326
Cycle TAE-C	✓	×	✓	0.9363	0.8673
DA-Former	✓	✓	✓	**0.9436**	**0.8721**

TABLE III
COMPLEXITY ANALYSIS OF DIFFERENT METHODS ON FARMLAND DATASETS

Methods	Params(K)	FLOPs(M)	Test time(s)
Re3FCN	464	**0.6**	0.026
CSANet	2427	140	0.035
SST-Former	2500	145	**0.02**
DA-Former	**398**	30	0.029

samples. Sharing the parameters of the dual-TAE middle layers increases Kappa by 1.11% for HSIs-CD, indicating that the shared middle layers can pull the two feature distributions into the same distribution, but its ability is limited. The Cycle TAE-C enhances kappa by 4.58% compared to the TAE-C, indicating that the cycle TAE can achieve domain alignment. Through the combination of two domain alignment methods (Shared TAE and Cycle TAE), domain alignment can be achieved. By combining two domain alignment methods (Shared TAE and Cycle TAE), also known as DA-Former, the accuracy of the CD is further increased and has the highest value. In conclusion, Shared and Cycle TAE-C can align the domains well. Moreover, combining autoencoders with domain alignment learning and limited samples can be used as a semi-supervised learning method for HSIs-CD.

E. Hyperparameter Analysis

The hyperparameters influence the performance of DA-Former significantly. Here, we examine the hyperparameter λ (L_{cd} weight) and patch size. The results are presented in Fig. 9.

We experiment with $\lambda = \{0.2, 0.3, 0.4, 0.5, 0.6, 0.7\}$ on three datasets to find the optimal L_{cd} weight. The experiments show that on Santa Barbara, the performance first increases and then decreases, with the highest Kappa provided for $\lambda = 0.4$. The Farmland data shows an "M" trend with the highest Kappa at $\lambda = 0.4$. The Bay Area data exhibits that 0.4 and 0.7 are the best λ. Regarding balance, 0.4 is the optimal L_{cd} weight.

Five patch sizes $\{3, 5, 7, 9, 11\}$ are analyzed on three datasets. Fig. 9(b) shows that Kappa increases and decreases for all three datasets as the patch size increases, and Kappa maximizes for a patch size equal to 9.

We chose 10, 50, 100, 200, 300 as the number of labeled samples to analyze the impact of training sample size and demonstrate the advantage of DA-Former in the case of limited labeled samples. Kappa initially increases as the number of training samples increases and then tends to level off, as shown in Fig. 9(c). When the number of training samples becomes greater than 100, Kappa tends to be saturated, indicating that DA-Former has little increase in Kappa under more labeled samples and has strong feature extraction capabilities in combination with unlabeled samples and limited labeled samples.

F. Complexity Analysis

Another key factor in measuring the impact of CD is the computational complexity of DL methods. Four DL methods are analyzed, including Re3FCN, CSANet, SST-Former, and DA-Former. We use parameters (Params), floating-point operations (FLOPs), and Test Time to analyze the complexity of these methods. The results are shown in Table III. They are all calculated under a pair of patches with a size of $9 \times 9 \times 154$, and in Re3FCN, there is a PCA that reduces the dimension from 154 to 30.

According to Table III, DA-Former has a smaller Params than other methods. The reason for the low FLOPs of Re3FCN may be due to the dimensionality reduction by CVA. The FLOPs of DA-Former is much smaller than that of SST-Former and CSANet. The test time of SST-Former is much shorter than others. On the contrary, the test time of CSANet on a pair of patches is 0.006 s longer than DA-Former. In conclusion, our DA-Former is at an advanced level in complexity and also has significant advantages over other methods in CD accuracy.

IV. CONCLUSION

Unsupervised CD algorithms have low accuracy, while supervised CD algorithms have high accuracy but require a significant amount of training samples; therefore, a semi-supervised algorithm with high accuracy and fewer training samples is needed. In this article, we propose a DA-Former algorithm to increase the accuracy of CD with small samples. First, a dual-TAE is designed to extract the bitemporal HSI features based on an unsupervised approach. Then, a cycle TAE structure between dual-TAE is used for domain alignment. Finally, a few training datasets are applied to separate the features of different objects and train the classifier for CD. Three HSI datasets were tested and demonstrated that the DA-Former can achieve the highest CD accuracy when few samples are available.

The proposed algorithm can be applied to emergent tasks with small sample sizes and high accuracy requirements, such as earthquakes. TAE can also be used in other unsupervised HSI tasks. Typically, our algorithm requires a training set of 100 pixels per class, but the number of training sets should also expand as the detection area expands accordingly; nevertheless, this method still requires a certain amount of human intervention, and thereby, algorithms with smaller sample sizes are worth investigating. Our future work will further reduce the sample size for CD based on few-shot learning.

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